

JOIcode Side-by-Side Report · gpt41_mini

Side-by-side GT vs Generated JOIcode report for **gpt41_mini**. This report is rendered from row-level benchmark outputs and is print-friendly for PDF export.

ROWS 3	AVG DET 69.8584	DET PASS 0.0%	GT EXACT 0.0%	AVG PROMPT TOKENS 24365.7
AVG COMPLETION TOKENS 86.0	AVG LLM LATENCY 3.263s	P50 LLM LATENCY 3.495s	PEAK VRAM 0.000 GB	OOM COUNT 0

RUN METADATA

Results Dir	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/results/gpt41_mini_retrieval_allcats_20260508/retrieval_gpt41_mini_cat4
Output Dir	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/results/gpt41_mini_retrieval_allcats_20260508/retrieval_gpt41_mini_cat4/side_by_side_report_gpt41_mini_failures_only
Suite	paper_with_cloud_ref
Genome	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/genomes/example_genome.json
Created At	2026-05-08T09:37:38.567763
Categories	4
Limit/Category	-
Rows	91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120

TOP FAILURE REASONS	
gt_mismatch	3
gt_receiver_coverage	3
gt_service_coverage	3
extraneous	2
semantic	2
enum_grounding	1

CONNECTION CATEGORIES
connections
categories

MODEL RUNTIME SNAPSHOT

MODEL GPT-4.1-mini	RESOLVED MODEL -	WORKER PYTHON -	SERVICE LIST service_list_ver2.0.1.json	CANDIDATE K 1
REPAIR ATTEMPTS 0	DET PROFILE openai			

ROW DETAILS

Row 106 · Category 4

When the entrance door lock is unlocked, turn on the living room light.

현관 도어락이 잠금해제되면 거실 조명을 켜줘.

GT name: WhenEntranceDoorLockIsUnlockedTurnOn				
Service list: service_list_ver2.0.1.json				
Output name: EntranceDoorUnlockTurnOnLivingRoomLight				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.4728	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 1.0000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 25029
COMPLETION TOKENS 62	TOTAL TOKENS 25091	LLM LATENCY 3.495s	PIPELINE 3.508s	TOK/S 7178.0861
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 3 connected-device category group(s) were injected.

CONTEXT MODE
connected_devices scope

SNIPPET SOURCE
connected_devices_only

RETRIEVAL
skipped_connected_devices_present

CONFIGURED TOP-K
10

EFFECTIVE TOP-K
-

INJECTED CATEGORIES
3

DoorLock

3 services · 1 values · 2 functions

CONNECTED

VALUES

DoorLock_DoorLockState
raw DoorLockState · arg - · ret ENUM · Allow for the control of a door lock

FUNCTIONS

DoorLock_Lock
raw Lock · arg - · ret VOID · Allow for the control of a door lock

DoorLock_Unlock

raw Unlock · arg - · ret VOID · Allow for the control of a door lock

Light

CONNECTED

15 services · 8 values · 7 functions

VALUES

Light_ColorMode

raw ColorMode · arg - · ret ENUM · A numerical representation of the brightness intensity

Light_CurrentBrightness

raw CurrentBrightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentColorTemperature

raw CurrentColorTemperature · arg - · ret INTEGER · A numerical representation of the brightness intensity

Light_CurrentHue

raw CurrentHue · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentRGB

raw CurrentRGB · arg - · ret STRING · A numerical representation of the brightness intensity

Light_CurrentSaturation

raw CurrentSaturation · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentX

raw CurrentX · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentY

raw CurrentY · arg - · ret DOUBLE · A numerical representation of the brightness intensity

FUNCTIONS

Light_MoveToBrightness

raw MoveToBrightness · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToColorTemperature

raw MoveToColorTemperature · arg INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHue

raw MoveToHue · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHueAndSaturation

raw MoveToHueAndSaturation · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToRGB

raw MoveToRGB · arg INTEGER | INTEGER | INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToSaturation

raw MoveToSaturation · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToXY

raw MoveToXY · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Switch

CONNECTED

4 services · 1 values · 3 functions

VALUES

Switch_Switch

raw Switch · arg - · ret BOOL · Allows for the control of a Switch device

FUNCTIONS

Switch_Off

raw Off · arg - · ret VOID · Allows for the control of a Switch device

Switch_On

raw On · arg - · ret VOID · Allows for the control of a Switch device

Switch_Toggle

raw Toggle · arg - · ret BOOL · Allows for the control of a Switch device

GT JOICODE

cron="" · period=0

```
wait until ((#Entrance #DoorLock).doorlock_doorlockstate == "unlocked")

(#LivingRoom #Light).switch_on()
```

GENERATED JOICODE

cron="" · period=0

```
prev := (#DoorLock).doorlock_doorlockstate

curr = (#DoorLock).doorlock_doorlockstate

if (prev != "closed" and curr == "closed") {

    (#Entrance #Light).levelcontrol_movetolevel(100, 0)

    delay(10 SEC)

    (#Entrance #Light).switch_off()

}

prev = curr
```

FAILURE REASONS

- extraneous
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage
- semantic

LINE DIFF

GT	Generated
wait until ((#Entrance #DoorLock).doorlock_doorlockstate == "unlocked")	prev := (#DoorLock).doorlock_doorlockstate
(#LivingRoom #Light).switch_on()	curr = (#DoorLock).doorlock_doorlockstate
	if (prev != "closed" and curr == "closed") {
	(#Entrance #Light).levelcontrol_movetolevel(100, 0)
	delay(10 SEC)
	(#Entrance #Light).switch_off()
	}

GT	Generated
	prev = curr

Row 107 · Category 4

When motion is detected in the bedroom, turn on the bedroom air conditioner.

침실에 움직임이 감지되면 침실 에어컨을 켜줘.

GT name: WhenMotionIsDetectedInBedroomTurnOn				
Service list: service_list_ver2.0.1.json				
Output name: BedroomMotionTurnOnAC				
Retrieval cats: not used (connected_devices scope)				
DET 69.7752	PASS No	GT EXACT No	SIMILARITY 0.4150	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 1.0000	NUMERIC 1.0000	ENUM 0.0000	PROMPT TOKENS 23335
COMPLETION TOKENS 91	TOTAL TOKENS 23426	LLM LATENCY 3.687s	PIPELINE 3.698s	TOK/S 6353.3304
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
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MotionSensor

1 services · 1 values · 0 functions

VALUES

MotionSensor_Motion

raw Motion · arg - · ret BOOL · Motion sensor device

FUNCTIONS

None

AirConditioner

4 services · 2 values · 2 functions

VALUES

AirConditioner_AirConditionerMode

raw AirConditionerMode · arg - · ret ENUM · Controls air conditioner mode and temperature settings

AirConditioner_TargetTemperature

raw TargetTemperature · arg - · ret DOUBLE · Controls air conditioner mode and temperature settings

FUNCTIONS

AirConditioner_SetAirConditionerMode

raw SetAirConditionerMode · arg ENUM · ret VOID · Controls air conditioner mode and temperature settings

AirConditioner_SetTargetTemperature

raw SetTargetTemperature · arg DOUBLE · ret VOID · Controls air conditioner mode and temperature settings

GT JOICODE

cron="" · period=0

```
wait until ((#Bedroom #MotionSensor).motionsensor_motion == true)

(#Bedroom #AirConditioner).airconditioner_setairconditionermode("auto")
```

GENERATED JOICODE

cron="" · period=100

```
triggered := false
if ((#Bedroom #MotionSensor).motionsensor_motion == true) {
  if (triggered == false) {
    (#Bedroom #AirConditioner).switch_on()
    triggered = true
  }
} else {
  triggered = false
}
```

FAILURE REASONS

- enum_grounding
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage
- semantic

LINE DIFF

GT	Generated
wait until ((#Bedroom #MotionSensor).motionsensor_motion == true)	triggered := false
	if ((#Bedroom #MotionSensor).motionsensor_motion == true) {
	{
(#Bedroom #AirConditioner).airconditioner_setairconditionermode("auto")	if (triggered == false) {
	(#Bedroom #AirConditioner).switch_on()
	triggered = true
	}
	} else {
	triggered = false
	}

Row 118 · Category 4

When any presence sensor in the hallway detects presence, set all hallway lights to purple.

복도에 재실 센서가 하나라도 감지되면 모든 복도 조명을 보라색으로 설정해줘.

GT name: WhenAnyPresenceSensorInHallwayDetectsPresenceSet				
Service list: service_list_ver2.0.1.json				
Output name: HallwayPresencePurpleLight				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.4991	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 1.0000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 24733
COMPLETION TOKENS 105	TOTAL TOKENS 24838	LLM LATENCY 2.607s	PIPELINE 2.618s	TOK/S 9526.3299
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 3 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 3
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PresenceSensor 1 services · 1 values · 0 functions VALUES PresenceSensor_Presence raw Presence · arg - · ret BOOL · The ability to see the current status of a presence sensor device FUNCTIONS None	CONNECTED
Light	CONNECTED

15 services · 8 values · 7 functions

VALUES

Light_ColorMode

raw ColorMode · arg - · ret ENUM · A numerical representation of the brightness intensity

Light_CurrentBrightness

raw CurrentBrightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentColorTemperature

raw CurrentColorTemperature · arg - · ret INTEGER · A numerical representation of the brightness intensity

Light_CurrentHue

raw CurrentHue · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentRGB

raw CurrentRGB · arg - · ret STRING · A numerical representation of the brightness intensity

Light_CurrentSaturation

raw CurrentSaturation · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentX

raw CurrentX · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentY

raw CurrentY · arg - · ret DOUBLE · A numerical representation of the brightness intensity

FUNCTIONS

Light_MoveToBrightness

raw MoveToBrightness · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToColorTemperature

raw MoveToColorTemperature · arg INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHue

raw MoveToHue · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHueAndSaturation

raw MoveToHueAndSaturation · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToRGB

raw MoveToRGB · arg INTEGER | INTEGER | INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToSaturation

raw MoveToSaturation · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToXY

raw MoveToXY · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

ColorControl

CONNECTED

2 services · 1 values · 1 functions

VALUES

ColorControl_Color

raw Color · arg - · ret STRING · Allows for control of a color changing device

FUNCTIONS

ColorControl_SetColor

raw SetColor · arg STRING · ret VOID · Allows for control of a color changing device

GT JOICODE

cron="" · period=0

```
wait until (all(#Hallway #PresenceSensor).presencesensor_presence ==| true)

all(#Hallway #Light).colorcontrol_setcolor("128,0,128")
```

GENERATED JOICODE

cron="" · period=100

```
triggered := false
if (all(#Hallway #PresenceSensor).presencesensor_presence ==| true) {
  if (triggered == false) {
    all(#Hallway #Light).light_movetorgb(128, 0, 128)
    triggered = true
  }
} else {
  triggered = false
}
```

FAILURE REASONS

- extraneous
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage

LINE DIFF

GT	Generated
wait until (all(#Hallway #PresenceSensor).presencesensor_presence == true)	triggered := false
	if (all(#Hallway #PresenceSensor).presencesensor_presence == true) {
all(#Hallway #Light).colorcontrol_setcolor("128,0,128")	if (triggered == false) {
	all(#Hallway #Light).light_movetorgb(128, 0, 128)
	triggered = true
	}
	} else {
	triggered = false
	}